

Homework Set HW1 due 1/28/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)

This problem is exactly like Problems 5 and 6 page 11 in Section 1.1.

Consider the two points $(2, -8)$ and $(8, 3)$.

The distance between them is: _____

The x co-ordinate of the midpoint of the line segment that joins them is: _____

The y co-ordinate of the midpoint of the line segment that joins them is: _____

2.(1 pt) This problem is exactly like Problems 13 and 18 page 11 in Section 1.1.

The equation $|5x - 5| = 15$ has two solutions.

The sum of those two solutions is _____

3.(1 pt) This problem is Problems 13 page 11 in Section 1.1.

The equation $|2x + 4| = 16$ has two solutions.

The sum of those two solutions is _____

4.(1 pt) This problem is Problems 14 page 11 in Section 1.1.

The equation $|5y + 2| = 12$ has two solutions.

The sum of those two solutions is _____

5.(1 pt) This problem is Problems 16 page 11 in Section 1.1.

The equation $|5 - 3t| = 14$ has two solutions.

The sum of those two solutions is _____

6.(1 pt) This problem is Problem 25 page 11 in Section 1.1.

Solve the following inequality. Write the answer in interval notation. **Note:** If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity*.

$$3x + 7 < 2$$

Answer: _____

7.(1 pt) This problem is Problem 26 page 11 in Section 1.1.

Solve the following inequality. Write the answer in interval notation. **Note:** If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity*.

$$5(3 - x) > 3x - 1$$

Answer: _____

8.(1 pt) This is problem 40, Section 1.1 page 11.

Find the center (h, k) and the radius r of the circle

$$4x^2 + 4y^2 + 4y - 15 = 0.$$

$h =$ _____

$k =$ _____

$r =$ _____

9.(1 pt) This is problem 42, Section 1.1 page 11.

Find the center (h, k) and the radius r of the circle

$$2x^2 + 2y^2 + 2x - 6y - 9 = 0.$$

$h =$ _____

$k =$ _____

$r =$ _____

10.(1 pt) This problem is like Problems 31 and 32 page 11 in Section 1.1.

Solve the following inequality. Write the answer in interval notation.

Note: If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity*.

$$x^2 + 9x + 18 > 0$$

Answer: _____

11.(1 pt) This is Problem 32 page 11 in Section 1.1.

Solve the following inequality. Write the answer in interval notation.

Note: If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity*.

$$s^2 + 3s - 4 > 0$$

Answer: _____

12.(1 pt) This is problem 44 in Section 1.1 page 11.

Use sum and difference formulas from trig to find the exact value

$$\cos\left(\frac{7\pi}{12}\right) =$$

13.(1 pt) This is problem 46 in Section 1.1 page 11.

Use sum and difference formulas from trig to find the exact value

$$\sin(165^\circ) =$$

14.(1 pt)

If $\sin(\theta) = -\frac{2}{4}$, and θ is in quadrant IV, then find

(a) $\cos(\theta) =$ _____

(b) $\tan(\theta) =$ _____

(c) $\sec(\theta) =$ _____

(d) $\csc(\theta) =$ _____

(e) $\cot(\theta) =$ _____

This problem corresponds to problems 43 - 46 in Section 1.1 page 11. You need to be able to do this type of problem.

15.(1 pt) Section 1.2, problem 10 page 17.

Find the equation of the line (in slope intercept form) with x-intercept $(4.5, 0)$ and y-intercept $(0, -5.4)$.

$y =$ _____

16.(1 pt) Section 1.2, problem 12 page 17.

Find the equation of the line (in slope intercept form) passing through the point $(4, 5)$ and parallel to the line passing through $(2, 1)$ and $(5, 9)$.

$y =$ _____

17.(1 pt) Section 1.2, problem 14 page 17.

Find the equation of the line (in slope intercept form) passing through the point $(-1,6)$ and perpendicular to the line passing through the origin with slope $.5$.

$y =$ _____

18.(1 pt) Section 1.2, problem 2 page 17.

Find the equation of the line (in slope intercept form) passing through the points $(1,4)$ and $(3,6)$.

$y =$ _____

19.(1 pt) Section 1.2, problem 4 page 17.

Find the equation of the horizontal line passing through the point $(-2, -5)$.

$y =$ _____

20.(1 pt) Find the equation of the line which bisects the line segment from $(0,0)$ to $(2,6)$ at right angles.

$y =$ _____

21.(1 pt) Find the equation of the line passing through the point $(7,9)$ with slope $1/3$.

$y =$ _____

22.(1 pt) Section 1.3 problem 2 page 31.

Given the function $f(x) = -x^2 + 2x + 3$. Calculate the following values:

$f(0) =$ _____

$f(1) =$ _____

$f(-2) =$ _____

The domain of the function $f(x) = -x^2 + 2x + 3$ is _____.

Write the answer in interval notation.

Note: If the answer includes more than one interval write the intervals separated by the union symbol, U. If needed enter $-\infty$ as $-infinity$ and ∞ as $infinity$.

23.(1 pt) Section 1.3 problem 7 page 31.

The domain of the function

$$\sqrt{x(x+2)}$$

is _____

Write the answer in interval notation.

Note: If the answer includes more than one interval write the intervals separated by the union symbol, U. If needed enter $-\infty$ as $-infinity$ and ∞ as $infinity$. Given the function $f(x) = \sqrt{x(x+2)}$. Calculate the following values (Note: if x is not in the domain type UD for undefined):

$f(-1) =$ _____

$f(1/2) =$ _____

$f(1) =$ _____

24.(1 pt) Section 1.3 problem 8 page 31.

The domain of the function

$$\sqrt{x^2 + 5x + 6}$$

is _____

Write the answer in interval notation.

Note: If the answer includes more than one interval write the intervals separated by the union symbol, U. If needed enter $-\infty$ as $-infinity$ and ∞ as $infinity$. Given the function

$f(x) = \sqrt{x^2 + 5x + 6}$. Calculate the following values (Note: if x is not in the domain type UD for undefined):

$f(0) =$ _____

$f(1) =$ _____

$f(-2) =$ _____

25.(1 pt) Given that $f(x) = 7x + 8$ and $g(x) = 7 - x^2$, calculate

(a) $f \circ g(x) =$ _____

(b) $g \circ f(x) =$ _____

26.(1 pt) This is a slightly restated version of problem 14, Section 1.3 page 31.

Given the function $f(x) = 2 - 5x$, calculate the following values:

$f(a) =$ _____

$f(a+h) =$ _____

$\frac{f(a+h) - f(a)}{h} =$ _____

27.(1 pt) This is a slightly restated version of problem 16, Section 1.3 page 31.

Given the function $f(x) = 2x + 3x^2$, calculate the following values:

$f(a) =$ _____

$f(a+h) =$ _____

$\frac{f(a+h) - f(a)}{h} =$ _____

28.(1 pt) This is a slightly restated version of problem 18, Section 1.3 page 31.

Given the function $f(x) = |x|$, calculate the following values (Note: Assume that $a > 1$ and $0 < h < 1$):

$f(a) =$ _____

$f(a+h) =$ _____

$\frac{f(a+h) - f(a)}{h} =$ _____

29.(1 pt) This is a slightly restated version of problem 20, Section 1.3 page 31.

Given the function $f(x) = \frac{(x+1)}{(x-1)}$, calculate the following values:

$f(a) =$ _____

$f(a+h) =$ _____

$\frac{f(a+h) - f(a)}{h} =$ _____

30.(1 pt) This is problem 36, Section 1.3 page 31.

Given that $f(x) = \sin(x)$ and $g(x) = 1 - x^2$, calculate

(a) $f \circ g(x) =$ _____

(b) $g \circ f(x) =$ _____

31.(1 pt) Enter a T or an F in each answer space below to indicate whether the corresponding statement is true or false. You must get all of the answers correct to receive credit.

___1. The ratio of two odd functions is odd

___2. The sum of an even and an odd function is usually neither even or odd, but it may be even.

___3. A function cannot be both even and odd.

___4. The product of two odd function is odd

- 5. The sum of two even functions is even
 —6. The composition of an even and an odd function is even
 —7. The composition of an odd function and an odd function is even
 —8. The product of two even functions is even

All of the answers must be correct before you get credit for the problem.

32.(1 pt) This is problem 12, Section 1.4 page 41.

Find the inverse for the function

$$f(x) = -3x + 2$$

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

33.(1 pt) This is problem 14, Section 1.4 page 41.

Find the inverse for the function

$$f(x) = x^2 - 5, x < 0$$

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

34.(1 pt) This is problem 16, Section 1.4 page 41.

Find the inverse for the function

$$G(x) = 10 - \sqrt{x}, x \geq 0$$

$$G^{-1}(x) = \underline{\hspace{2cm}}$$

35.(1 pt) Evaluate the following expressions.

$$\sin(\cos^{-1}(\frac{\sqrt{3}}{2})) \underline{\hspace{2cm}}$$

$$\tan(\cos^{-1}(\frac{1}{2})) \underline{\hspace{2cm}}$$

36.(1 pt) Evaluate the following expressions.

$$\sin(\cos^{-1}(\frac{\sqrt{2}}{2})) \underline{\hspace{2cm}}$$

$$\tan(\cos^{-1}(1)) \underline{\hspace{2cm}}$$

37.(1 pt)

Simplify the expression

$$\sin(2 \tan^{-1}(x)) = \underline{\hspace{2cm}}$$

38.(1 pt)

Simplify the expression

$$\cos(2 \sin^{-1}(x)) = \underline{\hspace{2cm}}$$